

Machine Learning

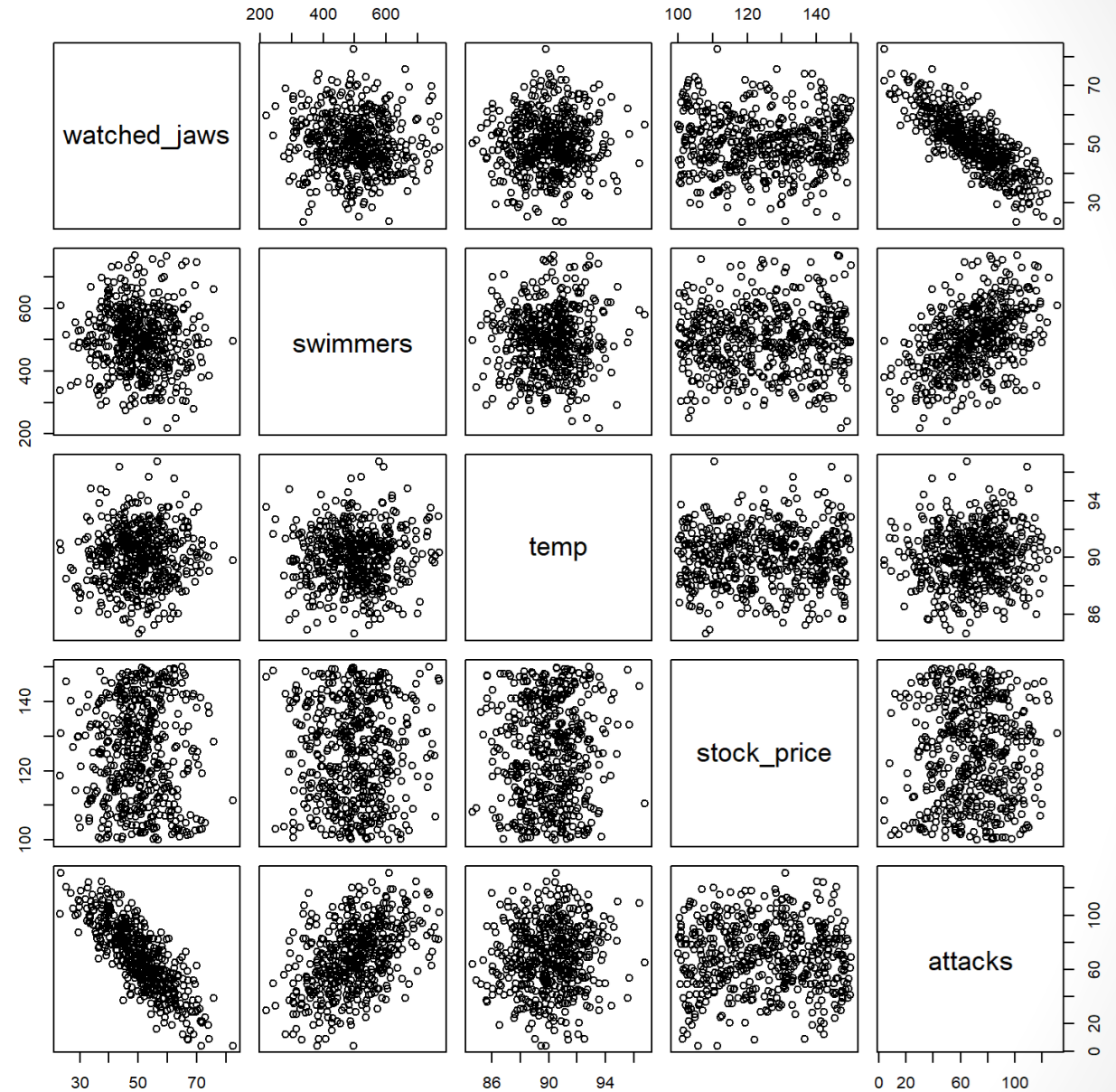
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Lecture 13: Manual Feature Selection

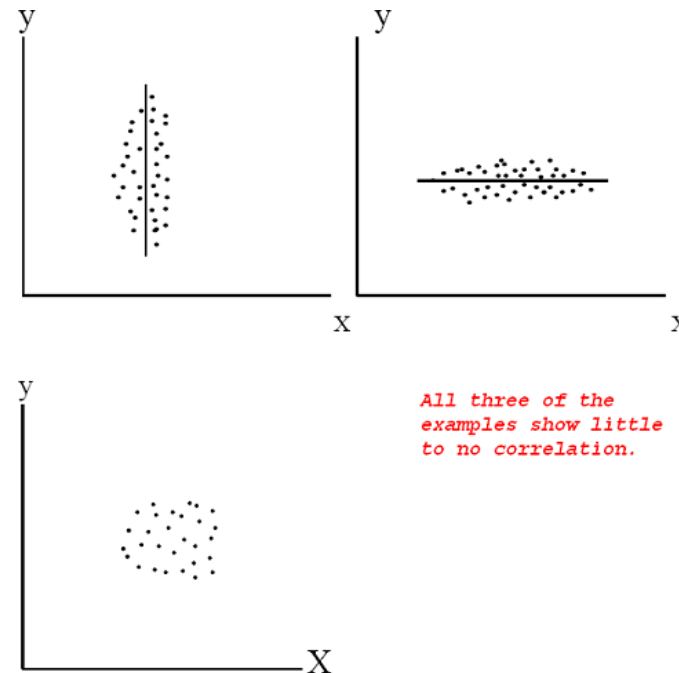
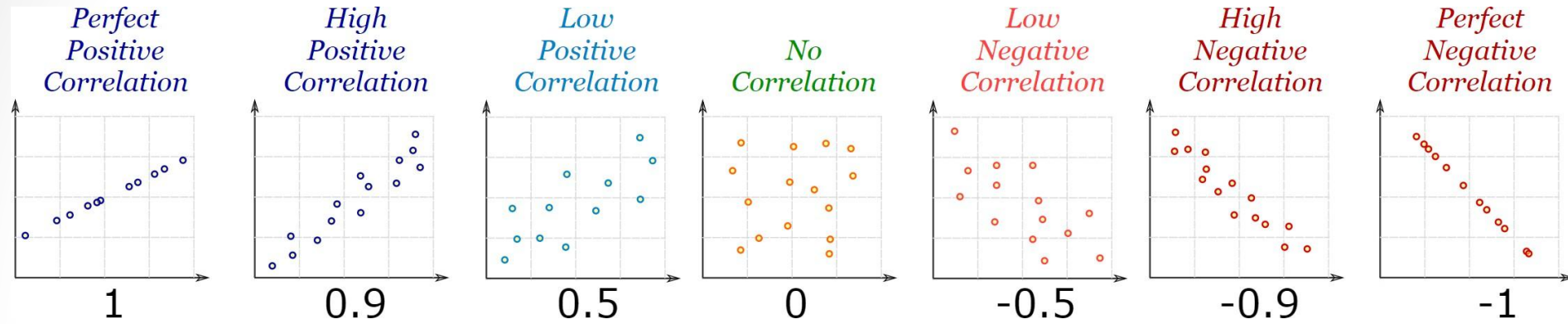
Manually select parameters

- Our data frame will consist of 1000 daily measurements of the following independent variables:
 - **attacks:** Number of shark attacks (output variable)
 - **swimmers:** Number of swimmers in water
 - **watched_jaws:** Percentage of swimmers who watched iconic Jaws movies
 - **temp:** Average temperature of the day
 - **stock_price:** The price of your favorite tech stock that day (a totally unrelated variable)

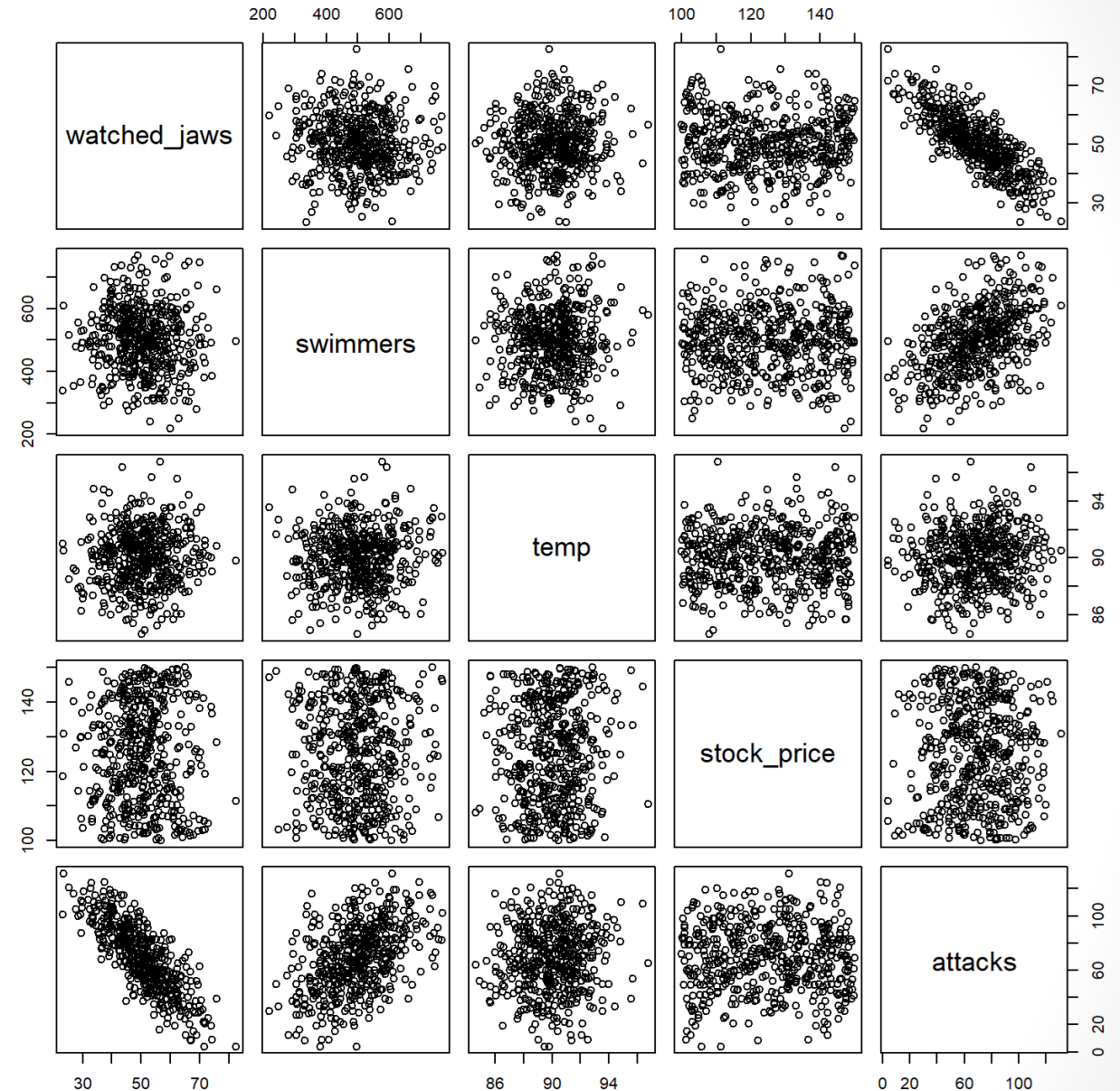
Scatter Diagrams



Scatter Diagrams



Scatter Diagrams



Regularization

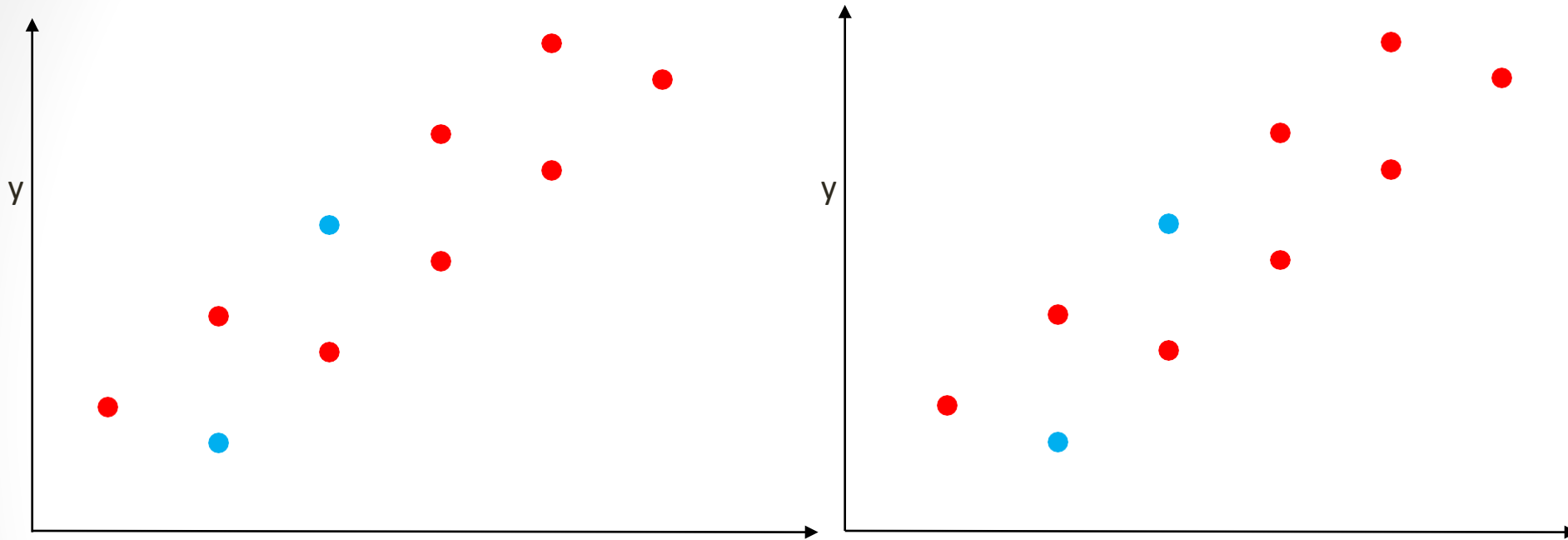
Regularization

- Regularization works on assumption that smaller weights generate simpler model and thus helps avoid overfitting.

$$f(x_i) = h_{\theta}x = \theta_0 + \theta_1x_1 + \theta_2x_2^2 + \theta_3x_3^3 + \theta_4x_4^4$$

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The fitting problem



$$h_{\theta} X = \Theta^T X = \theta_0 + \theta_1 x_1$$

$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2$$

$$\min_{\theta} \frac{1}{2m} \sum_{i=1}^m ((\theta_0 + \theta_1 x_1) - y_i)^2$$

Regularization

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$$J(\theta_0, \theta_1) = \frac{1}{2m} \sum_{i=1}^m (h_{\theta}(x_i) - y_i)^2$$

$$\min_{\theta} J(\theta) = \min_{\theta} \frac{1}{2m} \left[\sum_{i=1}^m ((\theta_0 + \theta_1 x_1) - y_i)^2 \right] + \lambda \sum_{j=1}^j \theta_j^2$$

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- λ is the penalty term or regularization parameter which determines how much to penalizes the weights
- When λ is zero then the regularization term becomes zero. We are back to the original Loss function
- When λ is large, we penalizes the weights and they become close to zero. This results is a very simple model having a high bias or is underfitting.
 - $h_{\theta} X = \theta_0$
- Use cross-validation to find the optimal value of λ

L2 Regularization or Ridge Regression

$$\min_{\theta} j(\theta) = \min_{\theta} \frac{1}{2m} \left[\sum_{i=1}^m ((\theta_0 + \theta_1 x_1) - y_i)^2 \right] + \lambda \sum_{j=1}^j \theta_j^2$$

- L2 regularization forces the weights to be small but does not make them zero and does not sparse solution.
- L2 is not robust to outliers as square terms blow up the error differences of the outliers and the regularization term tries to fix it by penalizing the weights
- Ridge regression performs better when all the input features influence the output and all with weights are of roughly equal size

L1 Regularization or Lasso Regression

$$\min_{\theta} J(\theta) = \min_{\theta} \frac{1}{2m} \left[\sum_{i=1}^m ((\theta_0 + \theta_1 x_1) - y_i)^2 \right] + \lambda \sum_{j=1}^j |\theta_j|$$

- L1 norm shrinks the parameters to zero.
 - When input features have weights closer to zero that leads to sparse L1 norm. In Sparse solution majority of the input features have zero weights and very few features have non-zero weights.
- Not all input features have the same influence on the prediction. L1 norm will assign a zero weight to features with less predictive power.
- L1 regularization does feature selection. It does this by assigning insignificant input features with zero weight and useful features with a non-zero weight.

Differences

$$\min_{\theta} J(\theta) = \min_{\theta} \frac{1}{2m} \left[\sum_{i=1}^m ((\theta_0 + \theta_1 x_1) - y_i)^2 \right] + \lambda \sum_{j=1}^j |\theta_j|$$

- **L1 Regularization**

- L1 penalizes sum of absolute value of weights.
- L1 has a sparse solution
- L1 has built in feature selection
- L1 is robust to outliers
- L1 generates model that are simple and interpretable but cannot learn complex patterns

- **L2 Regularization**

- L2 regularization penalizes sum of square weights.
- L2 has a non sparse solution
- L2 has no feature selection
- L2 is not robust to outliers
- L2 gives better prediction when output variable is a function of all input features
- L2 regularization is able to learn complex data patterns

- **Elastic net regularization is a combination of both L1 and L2 regularization**

Thank You